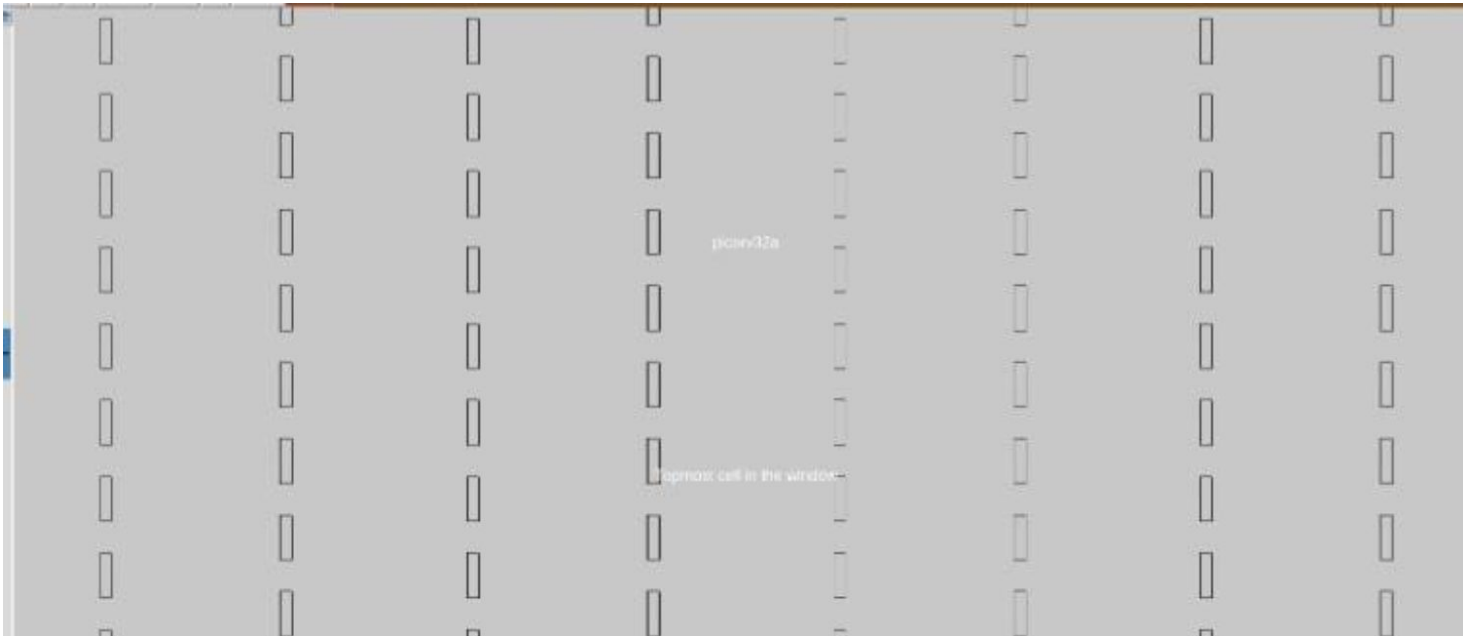
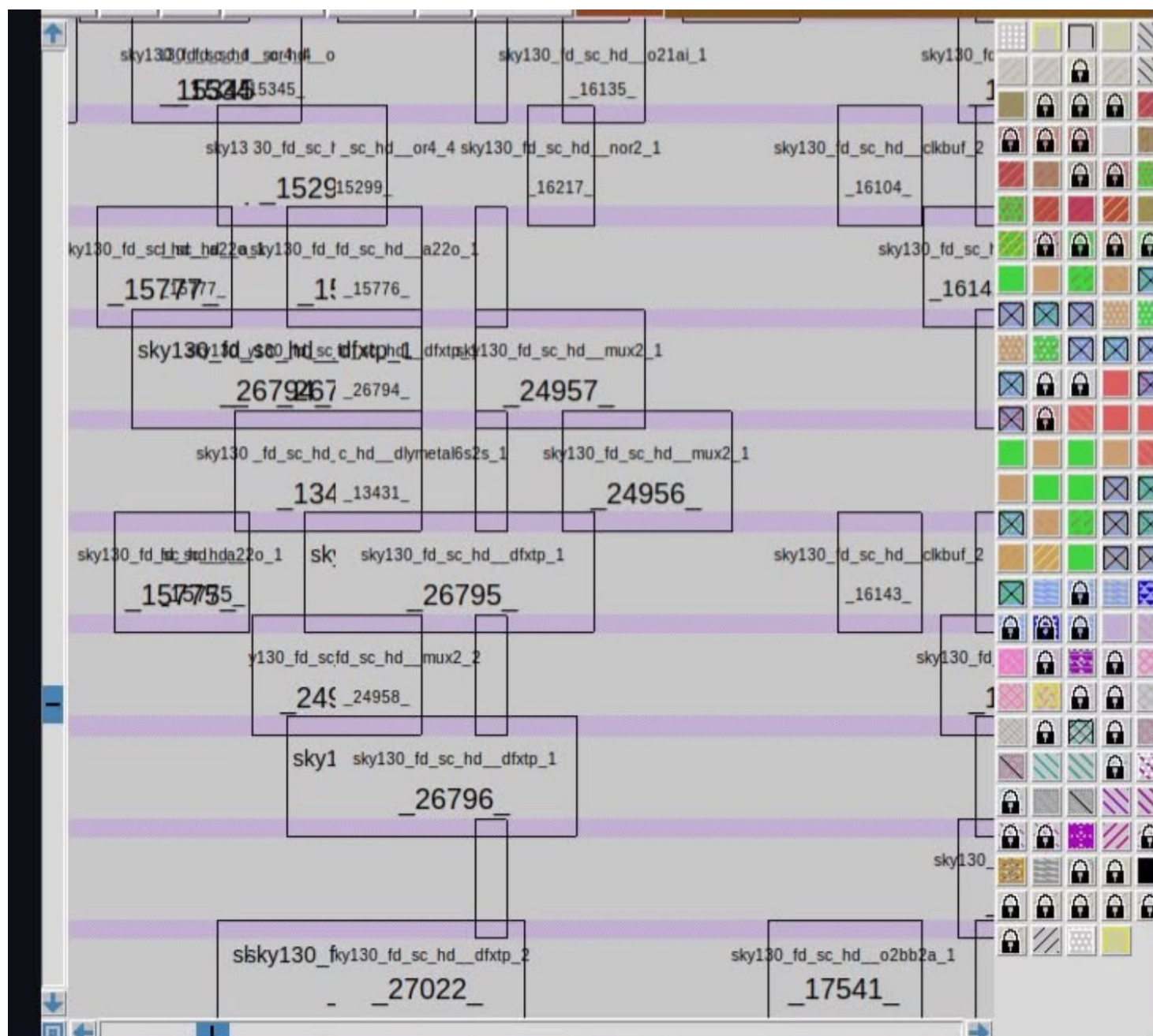
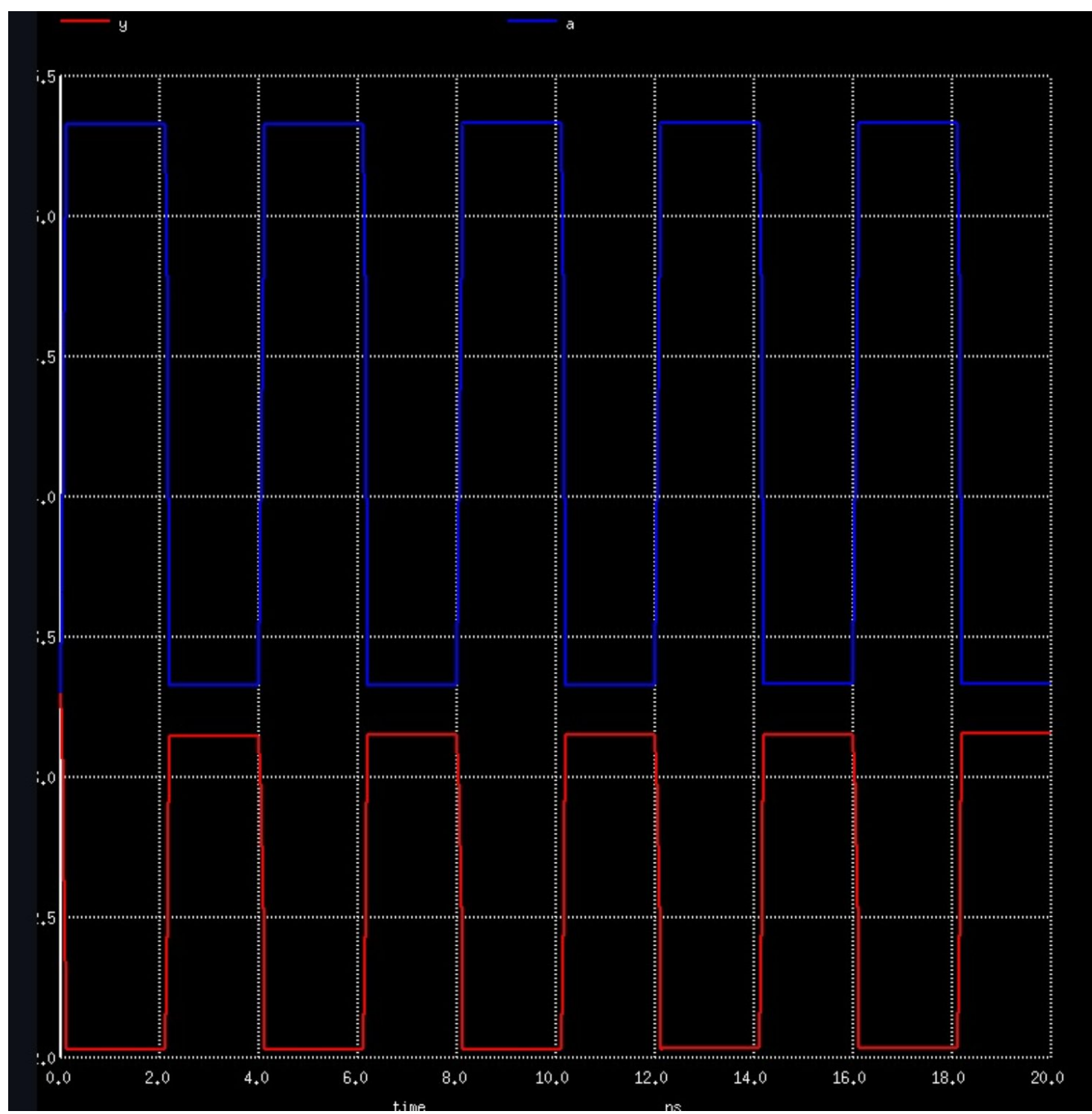


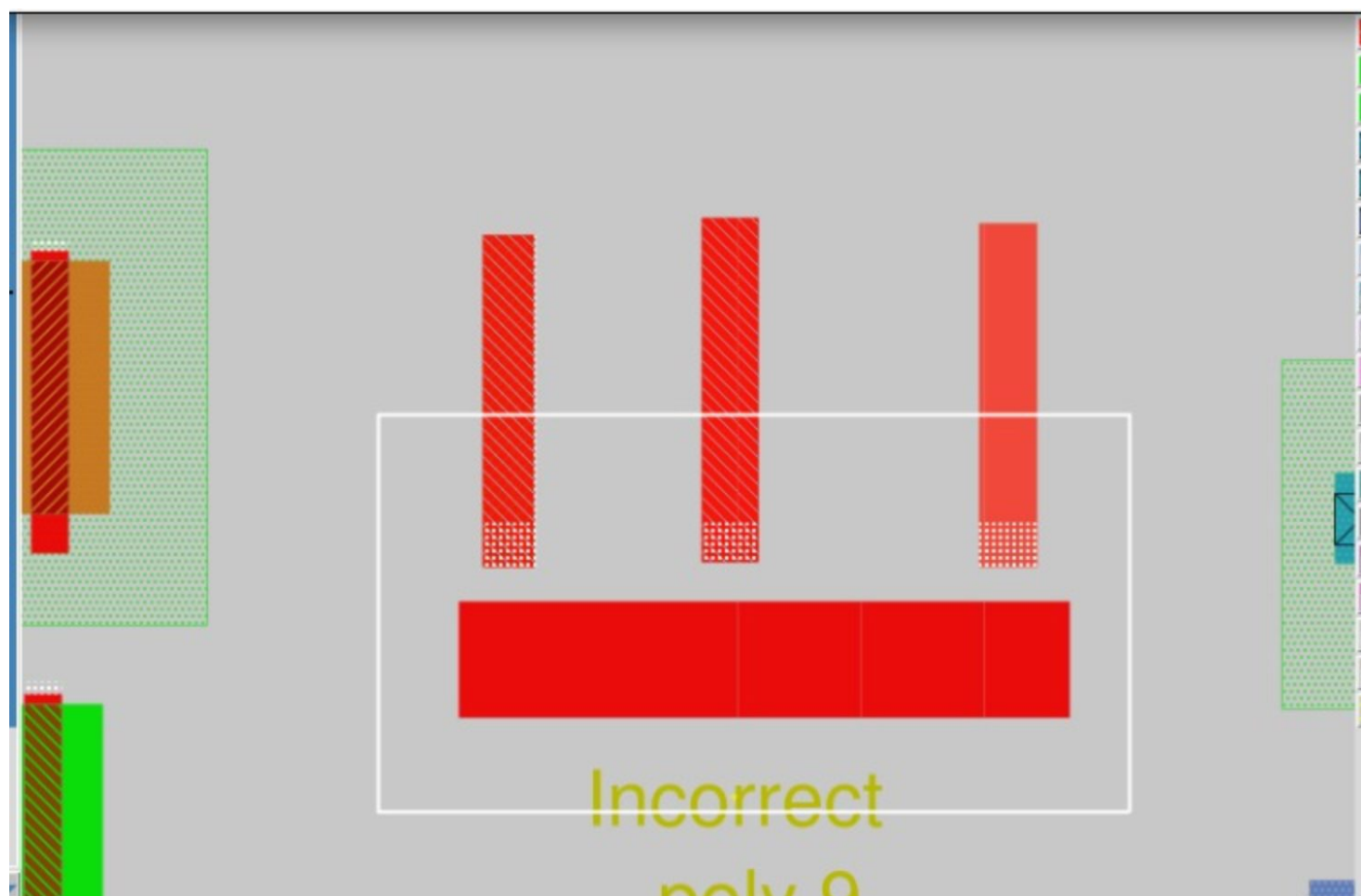
picov32a

Topmost cell in the window





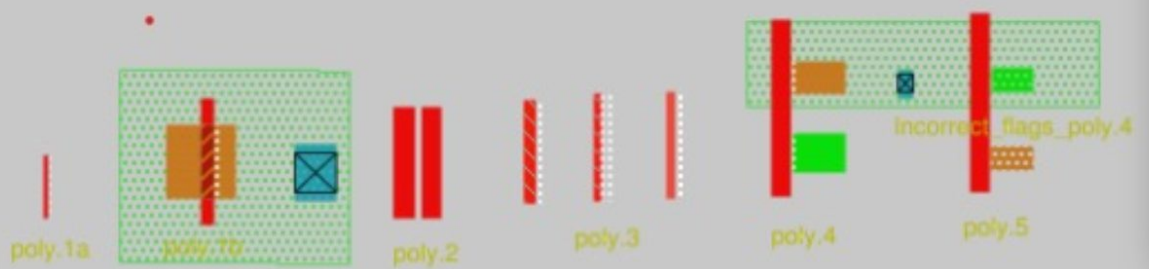




ERROR:_Incorrect_Implementation

nwell.4

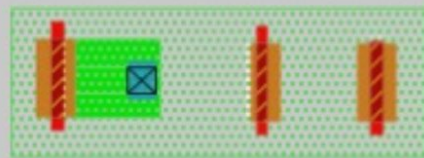
Poly



Correct_by_design

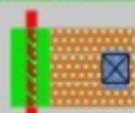
poly.12

poly.15



Not implemented

poly.16



poly.6



poly.7



poly.8



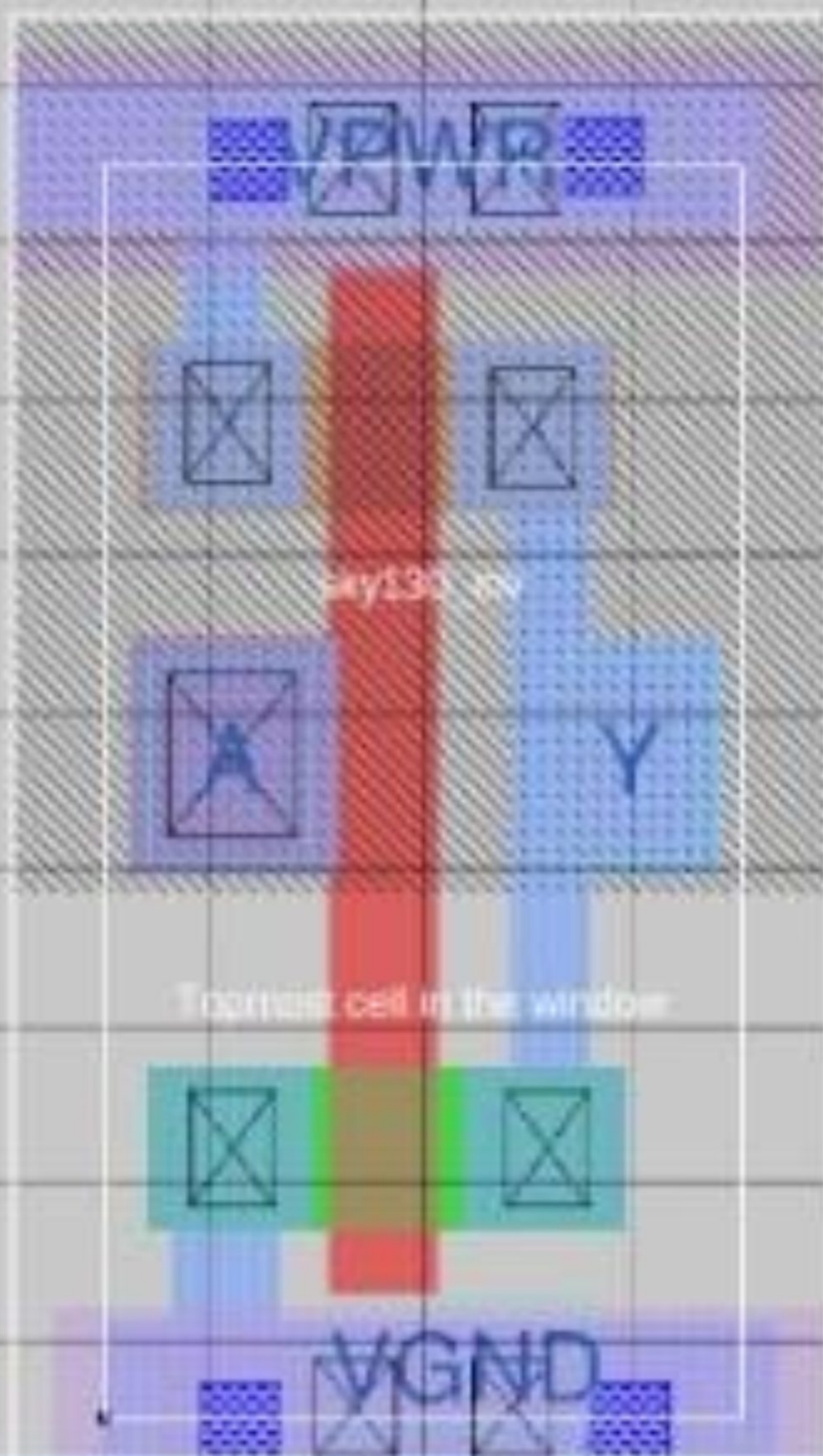
incorrect
poly.9

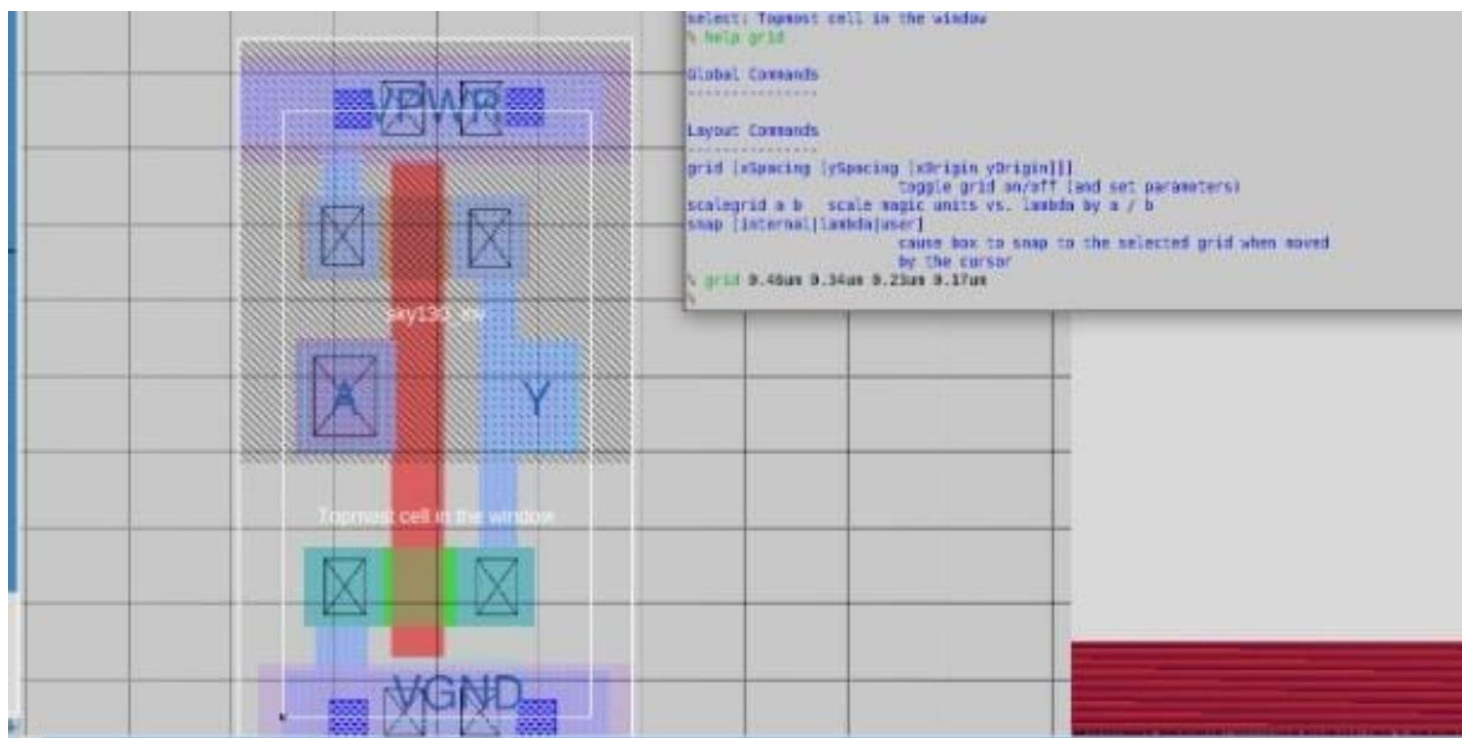


poly.10



poly.11





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Warning: /home/vsduser/Desktop/work/tools/openlane_working_dir/openlane/designs/picorv32a/src/sky130_fd_sc_hd_slow.lib line 24, default_fanout_load is 0.0.
 Warning: /home/vsduser/Desktop/work/tools/openlane_working_dir/openlane/designs/picorv32a/src/sky130_fd_sc_hd_fast.lib line 23, default_fanout_load is 0.0.

[INFO]: Setting output delay to: 2.4000000000000004

[INFO]: Setting input delay to: 2.4000000000000004

[INFO]: Setting load to: 0.01765

Startpoint: _31195_ (rising edge-triggered flip-flop clocked by clk)

Endpoint: _31195_ (rising edge-triggered flip-flop clocked by clk)

Path Group: clk

Path Type: min

Fanout	Cap	Slew	Delay	Time	Description

		0.00	0.00	0.00	clock clk (rise edge)
			0.00	0.00	clock network delay (ideal)
		0.00	0.00	0.00	^ _31195_/CLK (sky130_fd_sc_hd_dfxt2_2)
		0.03	0.18	0.18	^ _31195_/Q (sky130_fd_sc_hd_dfxt2_2)
2	0.00				cpuregs[0][0] (net)
		0.03	0.00	0.18	^ _18491_/A (sky130_fd_sc_hd_buf_1)
		0.02	0.04	0.23	^ _18491_/X (sky130_fd_sc_hd_buf_1)
1	0.00				_02835_ (net)
		0.02	0.00	0.23	^ _31195_/D (sky130_fd_sc_hd_dfxt2_2)
				0.23	data arrival time

		0.00	0.00	0.00	clock clk (rise edge)
			0.00	0.00	clock network delay (ideal)
			0.00	0.00	clock reconvergence pessimism
				0.00	^ _31195_/CLK (sky130_fd_sc_hd_dfxt2_2)
			-0.02	-0.02	library hold time
				-0.02	data required time

				-0.02	data required time
				-0.23	data arrival time

				0.24	slack (MET)

Startpoint: _32389_ (rising edge-triggered flip-flop clocked by clk)

Endpoint: _32288_ (rising edge-triggered flip-flop clocked by clk)

Startpoint: _32389_ (rising edge-triggered flip-flop clocked by clk)
 Endpoint: _32288_ (rising edge-triggered flip-flop clocked by clk)
 Path Group: clk
 Path Type: max

Fanout	Cap	Slew	Delay	Time	Description
		0.00	0.00	0.00	clock clk (rise edge)
			0.00	0.00	clock network delay (ideal)
		0.00	0.00	0.00	^ _32389_/CLK (sky130_fd_sc_hd__dfxtp_2)
4	0.01	0.13	0.67	0.67	^ _32389_/Q (sky130_fd_sc_hd__dfxtp_2)
					irq_mask[4] (net)
		0.13	0.00	0.67	^ _15411_/A (sky130_vsdinv)
2	0.01	0.09	0.13	0.80	v _15411_/Y (sky130_vsdinv)
					12736 (net)
		0.09	0.00	0.80	v _15412_/A (sky130_fd_sc_hd__nand2_2)
2	0.01	0.11	0.12	0.93	^ _15412_/Y (sky130_fd_sc_hd__nand2_2)
					12737 (net)
		0.11	0.00	0.93	^ _15417_/B1 (sky130_fd_sc_hd__o2111ai_2)
1	0.00	0.20	0.28	1.21	v _15417_/Y (sky130_fd_sc_hd__o2111ai_2)
					12742 (net)
		0.20	0.00	1.21	v _15426_/A (sky130_fd_sc_hd__nor2_2)
1	0.00	0.14	0.23	1.44	^ _15426_/Y (sky130_fd_sc_hd__nor2_2)
					12751 (net)
		0.14	0.00	1.44	^ _15443_/A (sky130_fd_sc_hd__nand3_2)
2	0.01	0.17	0.18	1.62	v _15443_/Y (sky130_fd_sc_hd__nand3_2)
					12768 (net)
		0.17	0.00	1.62	v _15446_/A2 (sky130_fd_sc_hd__o21bai_2)
6	0.02	0.41	0.40	2.02	^ _15446_/Y (sky130_fd_sc_hd__o21bai_2)
					12771 (net)
		0.41	0.00	2.02	^ _15447_/D1 (sky130_fd_sc_hd__o2111ai_2)
3	0.01	0.28	0.37	2.40	v _15447_/Y (sky130_fd_sc_hd__o2111ai_2)
					12772 (net)
		0.28	0.00	2.40	v _16145_/C (sky130_fd_sc_hd__nor3_2)
5	0.02	0.58	0.53	2.92	^ _16145_/Y (sky130_fd_sc_hd__nor3_2)
					13285 (net)
		0.58	0.00	2.92	^ _16146_/A (sky130_fd_sc_hd__nand3_2)
5	0.02	0.35	0.47	3.39	v _16146_/Y (sky130_fd_sc_hd__nand3_2)
					13286 (net)
		0.35	0.00	3.39	v _16147_/C (sky130_fd_sc_hd__nor3_2)
4	0.02	0.51	0.50	3.89	^ _16147_/Y (sky130_fd_sc_hd__nor3_2)
					13287 (net)
		0.51	0.00	3.89	^ _16148_/A (sky130_fd_sc_hd__nand3_2)
4	0.02	0.30	0.40	4.30	v _16148_/Y (sky130_fd_sc_hd__nand3_2)
					13288 (net)
		0.30	0.00	4.30	v _16149_/C (sky130_fd_sc_hd__nor3_2)
		0.62	0.57	4.87	^ _16149_/Y (sky130_fd_sc_hd__nor3_2)

		0.31	0.00	13.67	v	_16169_/C (sky130_fd_sc_hd__nor3_2)
		0.62	0.58	14.25	^	_16169_/Y (sky130_fd_sc_hd__nor3_2)
5	0.02					_13309_ (net)
		0.62	0.00	14.25	^	_16170_/A (sky130_fd_sc_hd__nand3_2)
4	0.02	0.31	0.45	14.70	v	_16170_/Y (sky130_fd_sc_hd__nand3_2)
						13310 (net)
		0.31	0.00	14.70	v	_16171_/C (sky130_fd_sc_hd__nor3_2)
5	0.02	0.63	0.58	15.27	^	_16171_/Y (sky130_fd_sc_hd__nor3_2)
						13311 (net)
		0.63	0.00	15.27	^	_16172_/A (sky130_fd_sc_hd__nand3_2)
5	0.02	0.32	0.45	15.73	v	_16172_/Y (sky130_fd_sc_hd__nand3_2)
						13312 (net)
		0.32	0.00	15.73	v	_16173_/C (sky130_fd_sc_hd__nor3_2)
3	0.01	0.46	0.45	16.18	^	_16173_/Y (sky130_fd_sc_hd__nor3_2)
						13313 (net)
		0.46	0.00	16.18	^	_16174_/A (sky130_fd_sc_hd__nand3_2)
4	0.01	0.26	0.36	16.54	v	_16174_/Y (sky130_fd_sc_hd__nand3_2)
						13314 (net)
		0.26	0.00	16.54	v	_16175_/A_N (sky130_fd_sc_hd__nand3b_2)
1	0.00	0.14	0.52	17.05	v	_16175_/Y (sky130_fd_sc_hd__nand3b_2)
						13315 (net)
		0.14	0.00	17.05	v	_16176_/B (sky130_fd_sc_hd__nor2_2)
1	0.00	0.14	0.17	17.22	^	_16176_/Y (sky130_fd_sc_hd__nor2_2)
						13316 (net)
		0.14	0.00	17.22	^	_16181_/A (sky130_fd_sc_hd__nor2_2)
1	0.00	0.04	0.09	17.31	v	_16181_/Y (sky130_fd_sc_hd__nor2_2)
						03928 (net)
		0.04	0.00	17.31	v	_32288_/D (sky130_fd_sc_hd__dfxtp_2)
				17.31		data arrival time
		0.00	12.00	12.00		clock clk (rise edge)
			0.00	12.00		clock network delay (ideal)
			0.00	12.00		clock reconvergence pessimism
				12.00	^	_32288_/CLK (sky130_fd_sc_hd__dfxtp_2)
		-0.28		11.72		library setup time
				11.72		data required time

				11.72		data required time
				-17.31		data arrival time

				-5.59		slack (VIOLATED)

tns -403.54

wns -5.59

% █

```
delta HPWL                2 %

[INFO DPL-0020] Mirrored 6036 instances
[INFO DPL-0021] HPWL before      863065.2 u
[INFO DPL-0022] HPWL after      848717.8 u
[INFO DPL-0023] HPWL delta      -1.7 %
[INFO]: Changing layout from /openLANE_flow/designs/picorv32a/runs/08-09_14-44/tmp/placement/8-re
sizer.def to /openLANE_flow/designs/picorv32a/runs/08-09_14-44/results/placement/picorv32a.placem
ent.def
[INFO]: Changing layout from /openLANE_flow/designs/picorv32a/runs/08-09_14-44/results/placement/
picorv32a.placement.def to /openLANE_flow/designs/picorv32a/runs/08-09_14-44/results/placement/pi
corv32a.placement.def
[INFO]: Taking a Screenshot of the Layout Using Klayout...
[INFO]: current step index: 12
Using Techfile: /home/vsduser/Desktop/work/tools/openlane_working_dir/pdks/sky130A/libs.tech/klay
out/sky130A.lyt
Using layout file: /openLANE_flow/designs/picorv32a/runs/08-09_14-44/results/placement/picorv32a.
placement.def
[INFO] Reading tech file: /home/vsduser/Desktop/work/tools/openlane_working_dir/pdks/sky130A/libs
.tech/klayout/sky130A.lyt
[INFO] Reading Layout file: /openLANE_flow/designs/picorv32a/runs/08-09_14-44/results/placement/p
icorv32a.placement.def
[INFO] Writing out PNG screenshot '/openLANE_flow/designs/picorv32a/runs/08-09_14-44/results/plac
ement/picorv32a.placement.def.png'
Done
[INFO]: Screenshot taken.
% set env::(SYNTH_MAX_FANOUT) 4
can't set "env::(SYNTH_MAX_FANOUT)": parent namespace doesn't exist
% set ::env(SYNTH_MAX_FANOUT) 4
4
% run synthesis
```

-0.2908	11.7092	library setup time
	11.7092	data required time

	11.7092	data required time
-16.3293		data arrival time

-4.6201		slack (VIOLATED)

```
% report_net -connections _04373_
```

```
Net _04373_
```

```
Driver pins
```

```
_22324_/Y output (sky130_fd_sc_hd__o21ai_2)
```

```
Load pins
```

```
_22325_/B input (sky130_fd_sc_hd__xor2_2)
```

```
_22330_/A1 input (sky130_fd_sc_hd__a21oi_2)
```

```
_22336_/A1 input (sky130_fd_sc_hd__a311o_2)
```

```
_22348_/A input (sky130_fd_sc_hd__nand3_2)
```

```
% replace_cell _22324_ sky130_fd_sc_hd__o21ai_4
```

```
1
```

```
% report_checks -fields {net cap slew input_pins} -digits 4
```

id	type	from	to	delay	setup	hold	path
3	0.02	0.12	0.00	12.33	^		_22380_/A2 (sky130_fd_sc_hd_o21ba1_2)
		0.20	0.22	12.55	v		_22380_/Y (sky130_fd_sc_hd_o21ba1_2)
							04413 (net)
2	0.01	0.20	0.00	12.55	v		_22385_/A (sky130_fd_sc_hd_nand2_2)
		0.09	0.16	12.70	^		_22385_/Y (sky130_fd_sc_hd_nand2_2)
							04416 (net)
1	0.00	0.09	0.00	12.70	^		_22390_/A (sky130_fd_sc_hd_and3_2)
		0.10	0.35	13.06	^		_22390_/X (sky130_fd_sc_hd_and3_2)
							04421 (net)
1	0.00	0.10	0.00	13.06	^		_22391_/B (sky130_fd_sc_hd_nor2_2)
		0.04	0.06	13.12	v		_22391_/Y (sky130_fd_sc_hd_nor2_2)
							01676 (net)
1	0.00	0.04	0.00	13.12	v		_33228_/A1 (sky130_fd_sc_hd_mux2_1)
		0.10	0.62	13.74	v		_33228_/X (sky130_fd_sc_hd_mux2_1)
							01677 (net)
1	0.00	0.10	0.00	13.74	v		_33682_/A1 (sky130_fd_sc_hd_mux4_1)
		0.14	1.10	14.83	v		_33682_/X (sky130_fd_sc_hd_mux4_1)
							01678 (net)
1	0.00	0.14	0.00	14.83	v		_33229_/A0 (sky130_fd_sc_hd_mux2_1)
		0.10	0.63	15.47	v		_33229_/X (sky130_fd_sc_hd_mux2_1)
							02529 (net)
1	0.00	0.10	0.00	15.47	v		_17453_/A (sky130_fd_sc_hd_or2b_2)
		0.10	0.65	16.12	v		_17453_/X (sky130_fd_sc_hd_or2b_2)
							14279 (net)
1	0.00	0.10	0.00	16.12	v		_17454_/C1 (sky130_fd_sc_hd_o211a_2)
		0.06	0.21	16.33	v		_17454_/X (sky130_fd_sc_hd_o211a_2)
							03989 (net)
		0.06	0.00	16.33	v		_35239_/D (sky130_fd_sc_hd_dfxtpt_2)
				16.33			data arrival time
		0.00	12.00	12.00			clock clk (rise edge)
			0.00	12.00			clock network delay (ideal)
			0.00	12.00			clock reconvergence pessimism
				12.00	^		_35239_/CLK (sky130_fd_sc_hd_dfxtpt_2)
			-0.29	11.71			library setup time
				11.71			data required time
				11.71			data required time
				-16.33			data arrival time
				-4.62			slack (VIOLATED)

tns -336.48

%

```

0.1123    6.7341 ^ _22349_/Y (sky130_fd_sc_hd__nand3b_2)
0.0788    6.8129 v _22351_/Y (sky130_fd_sc_hd__nand2_2)
0.0798    6.8927 ^ _22365_/Y (sky130_fd_sc_hd__nand3_4)
0.0747    6.9674 v _22380_/Y (sky130_fd_sc_hd__o21ba1_2)
0.0714    7.0388 ^ _22385_/Y (sky130_fd_sc_hd__nand2_1)
0.1338    7.1726 ^ _22390_/X (sky130_fd_sc_hd__and3_1)
0.0330    7.2056 v _22391_/Y (sky130_fd_sc_hd__nor2_1)
0.2707    7.4763 v _33228_/X (sky130_fd_sc_hd__mux2_1)
0.4759    7.9522 v _33682_/X (sky130_fd_sc_hd__mux4_1)
0.2774    8.2296 v _33229_/X (sky130_fd_sc_hd__mux2_1)
0.2121    8.4417 v _17453_/X (sky130_fd_sc_hd__or2b_1)
0.0965    8.5383 v _17454_/X (sky130_fd_sc_hd__o211a_1)
0.0000    8.5383 v _35239_/D (sky130_fd_sc_hd__dfxtp_1)
            8.5383 data arrival time

12.0000    12.0000 clock clk (rise edge)
0.0000    12.0000 clock source latency
0.0100    12.0100 ^ clk (in)
0.1016    12.1116 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
0.0697    12.1814 ^ clkbuf_1_0_0_clk/X (sky130_fd_sc_hd__clkbuf_1)
0.0910    12.2724 ^ clkbuf_1_0_1_clk/X (sky130_fd_sc_hd__clkbuf_1)
0.0811    12.3535 ^ clkbuf_2_1_0_clk/X (sky130_fd_sc_hd__clkbuf_1)
0.0911    12.4446 ^ clkbuf_2_1_1_clk/X (sky130_fd_sc_hd__clkbuf_1)
0.0987    12.5433 ^ clkbuf_3_2_0_clk/X (sky130_fd_sc_hd__clkbuf_1)
0.0988    12.6421 ^ clkbuf_4_5_0_clk/X (sky130_fd_sc_hd__clkbuf_1)
0.4392    13.0813 ^ clkbuf_5_10_0_clk/X (sky130_fd_sc_hd__clkbuf_1)
0.2471    13.3284 ^ clkbuf_leaf_190_clk/X (sky130_fd_sc_hd__clkbuf_16)
0.0000    13.3284 ^ _35239_/CLK (sky130_fd_sc_hd__dfxtp_1)
0.0000    13.3284 clock reconvergence pessimism
-0.1040    13.2244 library setup time
            13.2244 data required time
-----
            13.2244 data required time
            -8.5383 data arrival time
-----
            4.6862 slack (MET)

```

```

% echo $::env(CTS_CLK_BUFFER_LIST)
sky130_fd_sc_hd__clkbuf_1 sky130_fd_sc_hd__clkbuf_2 sky130_fd_sc_hd__clkbuf_4 sky130_fd_sc_hd__clkbuf_8
%
% set ::env(CTS_CLK_BUFFER_LIST) [lreplace $::env(CTS_CLK_BUFFER_LIST) 0 0]
sky130_fd_sc_hd__clkbuf_2 sky130_fd_sc_hd__clkbuf_4 sky130_fd_sc_hd__clkbuf_8
% echo $::env(CTS_CLK_BUFFER_LIST)
sky130_fd_sc_hd__clkbuf_2 sky130_fd_sc_hd__clkbuf_4 sky130_fd_sc_hd__clkbuf_8
% run_cts

```

```
INFO] Writing out PNG screenshot '/openLANE_flow/designs/picorv32a/runs/08-09_14-44/results/cts/
picorv32a.cts.def.png'
one
INFO]: Screenshot taken.
$ openroad
openROAD 0.9.0 1415572a73
This program is licensed under the BSD-3 license. See the LICENSE file for details.
Components of this program may be licensed under more restrictive licenses which must be honored.
$ read_lef /openLANE_flow/designs/picorv32a/runs/08-09_14-44/tmp/merged.lef
Notice 0: Reading LEF file: /openLANE_flow/designs/picorv32a/runs/08-09_14-44/tmp/merged.lef
Notice 0: Created 13 technology layers
Notice 0: Created 25 technology vias
Notice 0: Created 440 library cells
Notice 0: Finished LEF file: /openLANE_flow/designs/picorv32a/runs/08-09_14-44/tmp/merged.lef
$ read_def /openLANE_flow/designs/picorv32a/runs/08-09_14-44/results/cts/picorv32a.cts.def
Notice 0:
Reading DEF file: /openLANE_flow/designs/picorv32a/runs/08-09_14-44/results/cts/picorv32a.cts.def
Notice 0: Design: picorv32a
Notice 0: Created 411 pins.
Notice 0: Created 25636 components and 148154 component-terminals.
Notice 0: Created 2 special nets and 0 connections.
Notice 0: Created 19487 nets and 59928 connections.
Notice 0: Finished DEF file: /openLANE_flow/designs/picorv32a/runs/08-09_14-44/results/cts/picorv
32a.cts.def
$ write_db pico_cts.db
$ read_db pico_cts.db
$ read_verilog /openLANE_flow/designs/picorv32a/runs/08-09_14-44/results/synthesis/picorv32a.syn
thesis.cts.v
$ read_liberty -max $::env(LIB_SLOWEST)

$ read_liberty $::env(LIB_SYNTH_COMPLETE)

$ link_design picorv32a
WARNING ORD-1000] LEF master sky130_fd_sc_hd__tapvpwrvrgnd_1 has no liberty cell.
$ read_sdc /openLANE_flow/designs/picorv32a/src/base.sdc
INFO]: Setting output delay to: 2.4000000000000004
INFO]: Setting input delay to: 2.4000000000000004
INFO]: Setting load to: 0.01765
$ set_propagated_clock [all_clocks]
$ report_checks -path_delay min_max -format full_clock_expanded -digits 4
Startpoint: _33887_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _33852_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min
```

```

0.0913    0.2849 ^ clkbuf_1_1_1_clk/X (sky130_fd_sc_hd__clkbuf_2)
0.0851    0.3700 ^ clkbuf_2_3_0_clk/X (sky130_fd_sc_hd__clkbuf_2)
0.0913    0.4613 ^ clkbuf_2_3_1_clk/X (sky130_fd_sc_hd__clkbuf_2)
0.0953    0.5566 ^ clkbuf_3_6_0_clk/X (sky130_fd_sc_hd__clkbuf_2)
0.0953    0.6520 ^ clkbuf_4_12_0_clk/X (sky130_fd_sc_hd__clkbuf_2)
0.3937    1.0457 ^ clkbuf_5_25_0_clk/X (sky130_fd_sc_hd__clkbuf_2)
0.2401    1.2858 ^ clkbuf_leaf_200_clk/X (sky130_fd_sc_hd__clkbuf_16)
0.0000    1.2858 ^ _33852_/CLK (sky130_fd_sc_hd__dfxtp_1)
0.0000    1.2858 clock reconvergence pessimism
-0.0271   1.2587 library hold time
          1.2587 data required time
-----
          1.2587 data required time
          -1.4181 data arrival time
-----
          0.1594 slack (MET)

```

```

0.0000      8.2550  ^ _35239_/CLK (sky130_fd_sc_hd__dfxtp_1)
               data arrival time

12.0000      12.0000  clock clk (rise edge)
0.0000      12.0000  clock source latency
0.0100      12.0100  ^ clk (in)
0.1019      12.1119  ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
0.0817      12.1936  ^ clkbuf_1_0_0_clk/X (sky130_fd_sc_hd__clkbuf_2)
0.0913      12.2849  ^ clkbuf_1_0_1_clk/X (sky130_fd_sc_hd__clkbuf_2)
0.0851      12.3700  ^ clkbuf_2_1_0_clk/X (sky130_fd_sc_hd__clkbuf_2)
0.0913      12.4613  ^ clkbuf_2_1_1_clk/X (sky130_fd_sc_hd__clkbuf_2)
0.0953      12.5566  ^ clkbuf_3_2_0_clk/X (sky130_fd_sc_hd__clkbuf_2)
0.0953      12.6520  ^ clkbuf_4_5_0_clk/X (sky130_fd_sc_hd__clkbuf_2)
0.2516      12.9035  ^ clkbuf_5_10_0_clk/X (sky130_fd_sc_hd__clkbuf_2)
0.1928      13.0963  ^ clkbuf_leaf_190_clk/X (sky130_fd_sc_hd__clkbuf_16)
0.0000      13.0963  ^ _35239_/CLK (sky130_fd_sc_hd__dfxtp_1)
0.0000      13.0963  clock reconvergence pessimism
-0.1060      12.9904  library setup time
               12.9904  data required time
-----
               12.9904  data required time
               -8.2550  data arrival time
-----
               4.7354  slack (MET)

```

12.9904 data required time

12.9904 data required time

-8.2550 data arrival time

4.7354 slack (MET)

% check_clock_skew

Invalid command name "check_clock_skew"

% check_clock_skew -hold

Invalid command name "check_clock_skew"

% report_clock_skew -hold

clock clk

Latency	CRPR	Skew
---------	------	------

34515/CLK ^		
---------------	--	--

1.34		
------	--	--

33992/CLK ^		
---------------	--	--

0.80	0.00	0.54
------	------	------

% check_clock_skew -setup

Invalid command name "check_clock_skew"

% report_clock_skew -setup

clock clk

Latency	CRPR	Skew
---------	------	------

34515/CLK ^		
---------------	--	--

1.34		
------	--	--

33992/CLK ^		
---------------	--	--

0.80	0.00	0.54
------	------	------

